

NIRAV SURABHI

Half on a **screen**, half on a **workbench** — software + mechanical engineering.

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github.com/21J3phy

EDUCATION

- Purdue University**
B.S. Mechanical Engineering
Independence High School
Loudoun County, VA · Senior
Coursework: AP Computer Science A, AP Physics C, Multivariable Calculus.
- FALL 2026 · INCOMING
- Central Virginia Governor's School**
Science & Technology · Mathematics scholar
- JUN-JUL 2024

SOFTWARE — FOUNDER PROJECTS

- Chad vs. Chud**
chadvschud.com
Real-time 1v1 webcam face-rating battle scored by an AI. Built the Next.js client, a Socket.IO/WebRTC matchmaking service, and a Python/FastAPI inference service (ResNet-18 + ArcFace + LLM scorer). **10,000+ users, 100+ concurrent.**
Next.js · TypeScript · FastAPI · WebRTC · PyTorch · Supabase · Fly.io
- 2026-NOW · CO-FOUNDER/CEO
- b2alpha.io**
b2alpha.io
Open agent-to-agent protocol — secure messaging + DNS integration; prompt engineering for structured, parseable output.
2,000+ business users.
TypeScript · Node.js · OpenAI API · Vercel · Supabase · DNS
- 2026-NOW · CO-FOUNDER/CEO
- PredictionView**
predictionview.app
Full-stack Kalshi prediction-market trading hub — charts, an OpenAI analyst with web research, alerts, watchlists, a journal/backtester. Own the Express + Prisma/Postgres backend (127 routes, cron jobs) and Vitest tests.
React · TypeScript · Vite · Express · Prisma · PostgreSQL · OpenAI
- 2026-NOW · CO-FOUNDER/CEO
- Khali**
khali.info
Swipe-through news app — each story a short video card with a claim + agree/disagree and an OpenAI "ask this story" assistant. Native SwiftUI iOS + Expo/RN app on a Node + Python backend (ingest, clustering, clips/TTS). Submitted to YC Spring 2026.
Swift/SwiftUI · React Native · Expo · Node.js · Python · OpenAI · Supabase
- 2025-2026 · CO-FOUNDER/CEO

SKILLS

- LANGUAGES**
Java, Python, JavaScript, TypeScript, HTML, CSS, Git
- AI / ML**
Prompt engineering, LLMs, CNNs, computer vision, agent architecture
- CLOUD**
Fly.io, Vercel, Supabase, Fusion 360
- HARDWARE**
Arduino, load cells, soldering, multimeter, PID control

MECHANICAL / HARDWARE

- Passive Vortex Propulsion**
2024-2025 · 1ST AUTHOR
Built a wind tunnel from scratch; soldered & calibrated load cells with an Arduino; ran CFD in Fusion 360. **First demonstration of passive vortex propulsion in air** — published first-author paper; AIAA award & NASA Goddard invitation.
Arduino · C++ · load cells · Fusion 360 · CFD · Python
- FTC Robotics — Absolute Zero (12096)**
2022-2026 · CAPTAIN
Four years as captain & lead programmer; CADded and built every part in Fusion 360. Wrote autonomous & tele-op in Java — PID loops, sensor fusion, odometry. Reached the Chesapeake championship.
- EV & Salinity Sensor**
SCIENCE OLYMPIAD
Autonomous EV that accelerates then brakes to a target distance (Arduino, goBILDA motor, quadrature encoder, PID); Arduino salt-water salinity sensor reading conductivity.

EXPERIENCE

- AI Engineer & SWE Intern**
FEB 2021-NOW
Ember Learning (formerly DeAP Learning Labs)
Wrote the LLM prompts behind AP-tutor chatbots; built automated AI grading for AP DBQs/FRQs against the official rubric; indexed lecture transcripts in a vector DB for timestamp search; led the company website. **Team landed a Microsoft & OpenAI grant; \$1M+ valuation.**
- CAD Intern**
JUN 2025-MAY 2026
I3 Simulations
3D models and haptic-feedback prototypes for medical-training simulators.
- Bike Mechanic**
JUN 2021-AUG 2025
The Bike Lane · intern → part-time
Built bikes from the frame up; ran final safety checks.

HONORS & MUSIC

- AIAA Aerospace Award (RSEF special honor → NASA Goddard invitation)
- Captain, Science Olympiad
- Presidential Volunteer Service Award (Silver)
- Central Virginia Governor's School for Science & Technology — Mathematics, 2024
- 1st place, Agaudo guitar competition · All-Virginia Guitar Ensemble